News Topic Classification Using Machine Learning and BERT

Course Project — Natural Language Processing

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Overview

- This project compares traditional TF-IDF-based machine learning models with a fine-tuned BERT model on the AG News dataset.
- The main goal was not only to measure whether BERT performs better, but also to understand where that improvement appears. In particular, we wanted to see whether BERT helps most on news examples where the topic is not obvious from keywords alone.

Project Background

- **What is News Topic Classification?**
- News topic classification is a Natural Language Processing task.
- The goal is to automatically assign a topic label to a news article headline or short description.
- Example categories:
  - World
  - Sports
  - Business
  - Science / Technology
- This helps:
  - organize articles
  - improve search systems
  - build recommendation engines
  - analyze trends in media
- News topic classification is one of the most common tasks in NLP. A computer reads text and predicts which category it belongs to. This has practical value because large news companies publish huge amounts of content every day, and automatic categorization saves time while improving organization and recommendation quality.

Research Question

Our Main Question

- **We wanted to answer:**
- **How much does BERT improve topic classification compared with traditional TF-IDF-based machine learning models?**

Also:

- Is BERT's improvement large?
- Where exactly does BERT help?
- What kinds of mistakes does BERT reduce?
- Does BERT mainly help when keywords are misleading?

Our project was not simply about proving BERT is better. Many people already know transformer models perform strongly. Our deeper question was understanding exactly where BERT gains its advantage and whether that advantage comes from context understanding rather than keyword matching.

Dataset

- We used the **AG News Dataset**, a common benchmark dataset for text classification. Each example is a short news headline or description labeled as one of four topics.
- Contains **4 categories**:

Label ID	Category
0	World
1	Sports
2	Business
3	Sci/Tech

- AG News is a standard benchmark for text classification research. It contains short news headlines and short descriptions labeled into four categories. Because it is balanced and widely used, it provides a fair benchmark for comparing models.

Data Split

The original AG News training set contains 120,000 examples. We split it into 108,000 training examples and 12,000 validation examples, which kept the class distribution balanced across the split.

The validation set was used for model selection and hyperparameter tuning. The official 7,600-example AG News test set was kept separate during development and used only for the final evaluation.

Separating validation and test sets is important because it prevents us from accidentally optimizing directly on test data. That keeps our final evaluation fair and realistic.

Models Compared

We compared three traditional models with one transformer-based model.

Traditional Machine Learning Models

For the traditional models, text was converted into **TF-IDF unigram and bigram features**. These features were used to train:

- Most Frequent Baseline
- Multinomial Naive Bayes
- Logistic Regression

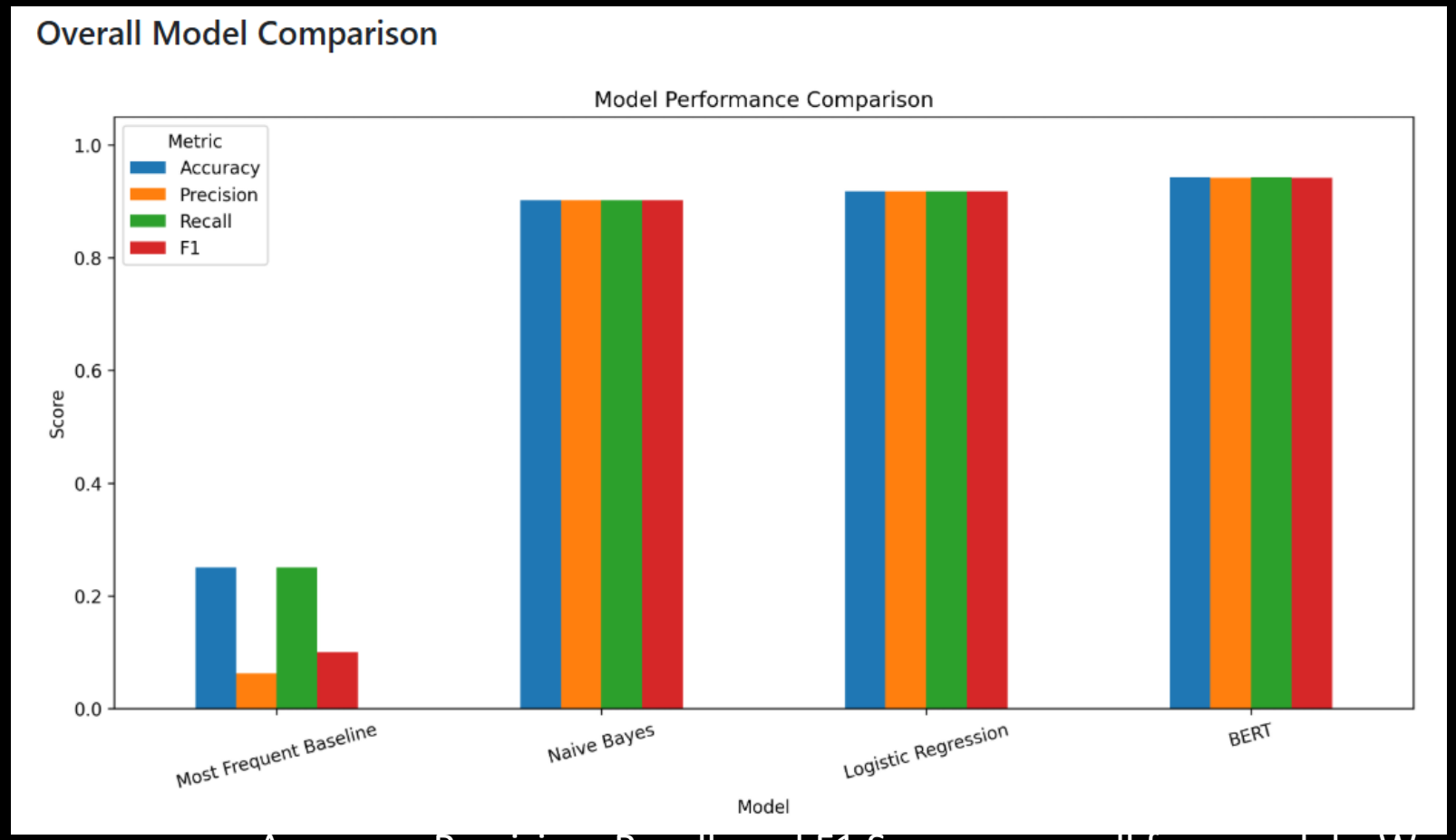
The most frequent baseline gives a simple lower-bound comparison, while Naive Bayes and Logistic Regression are common starting points for text classification.

BERT Model

- We fine-tuned bert-base-uncased for four-class classification. BERT represents tokens based on surrounding context, which can help when news topics overlap or when individual keywords are misleading.
- Due to time and compute limits, we trained one BERT configuration rather than running a large hyperparameter search.

We intentionally included both weak and strong baselines. This helps us understand whether BERT is only better than simple models, or whether it also clearly beats strong classical machine learning systems.

Performance Comparison



This figure compares Accuracy, Precision, Recall, and F1 Score across all four models. We can clearly see that Naive Bayes and Logistic Regression already perform strongly, both above 90% accuracy, while BERT achieves the best performance in every metric. Although the gap between Logistic Regression and BERT is not huge, BERT consistently leads across all evaluation measures, making it the strongest overall model in our project.

Final Results

All scores below were evaluated on the held-out AG News test set.

Model	Accuracy	Precision	Recall	F1 Score
Most Frequent Baseline	0.2500	0.0625	0.2500	0.1000
Naive Bayes	0.9024	0.9019	0.9024	0.9019
Logistic Regression	0.9180	0.9178	0.9180	0.9178
BERT	**0.9425**	**0.9424**	**0.9425**	**0.9424**

BERT achieved the highest score across all metrics. At the same time, Logistic Regression was a strong baseline, reaching over 91% accuracy with a simpler and cheaper model.

Error Analysis: BERT vs. Logistic Regression

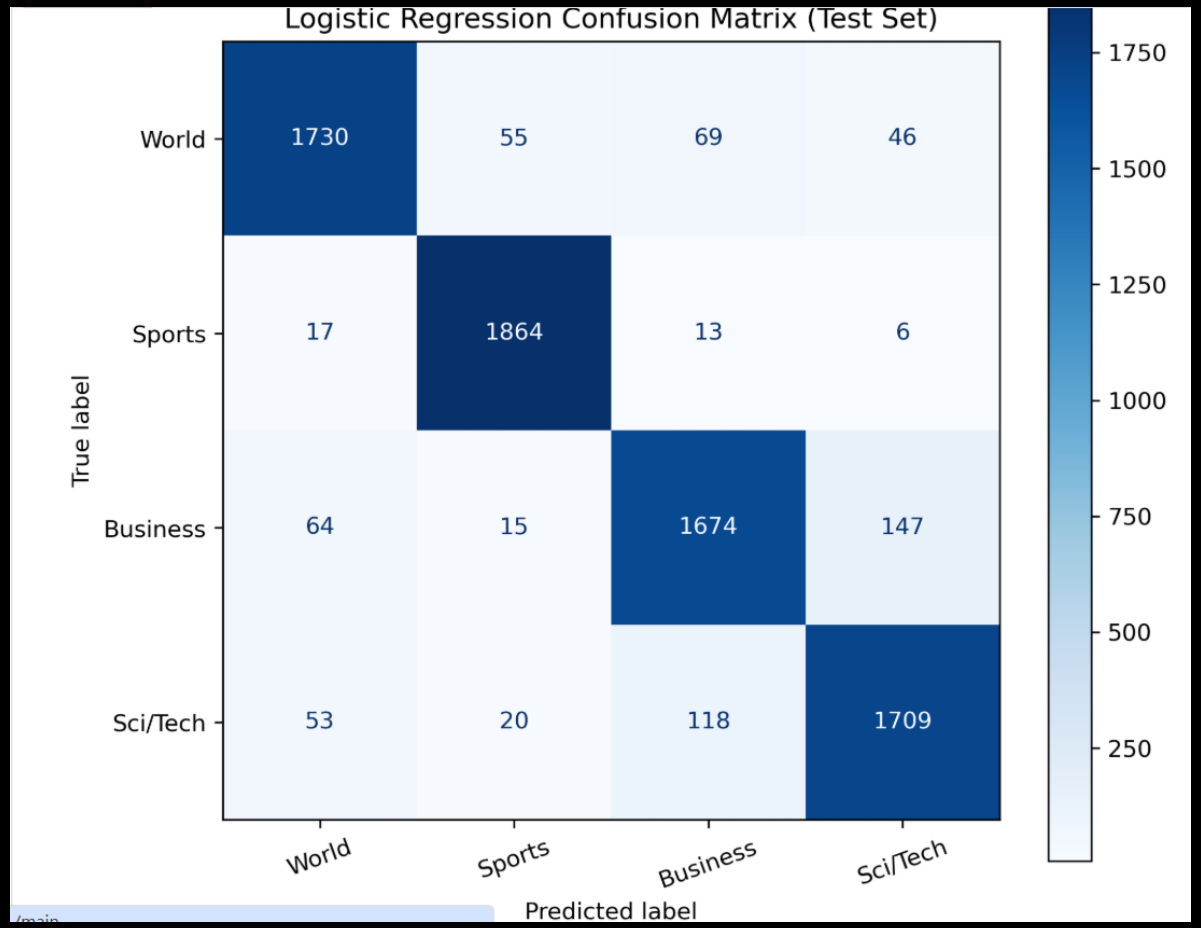
To understand the difference between Logistic Regression and BERT more directly, we compared their predictions on the same 7,600 test examples.

Case	Count	Share of Test Set
Both correct	6,862	90.3%
Logistic Regression wrong / BERT correct	301	4.0%
BERT wrong / Logistic Regression correct	115	1.5%
Both wrong	322	4.2%

BERT corrected 301 examples that Logistic Regression missed, while Logistic Regression corrected 115 examples that BERT missed. That gives BERT a net gain of 186 examples, which accounts for most of the accuracy gap between the two models.

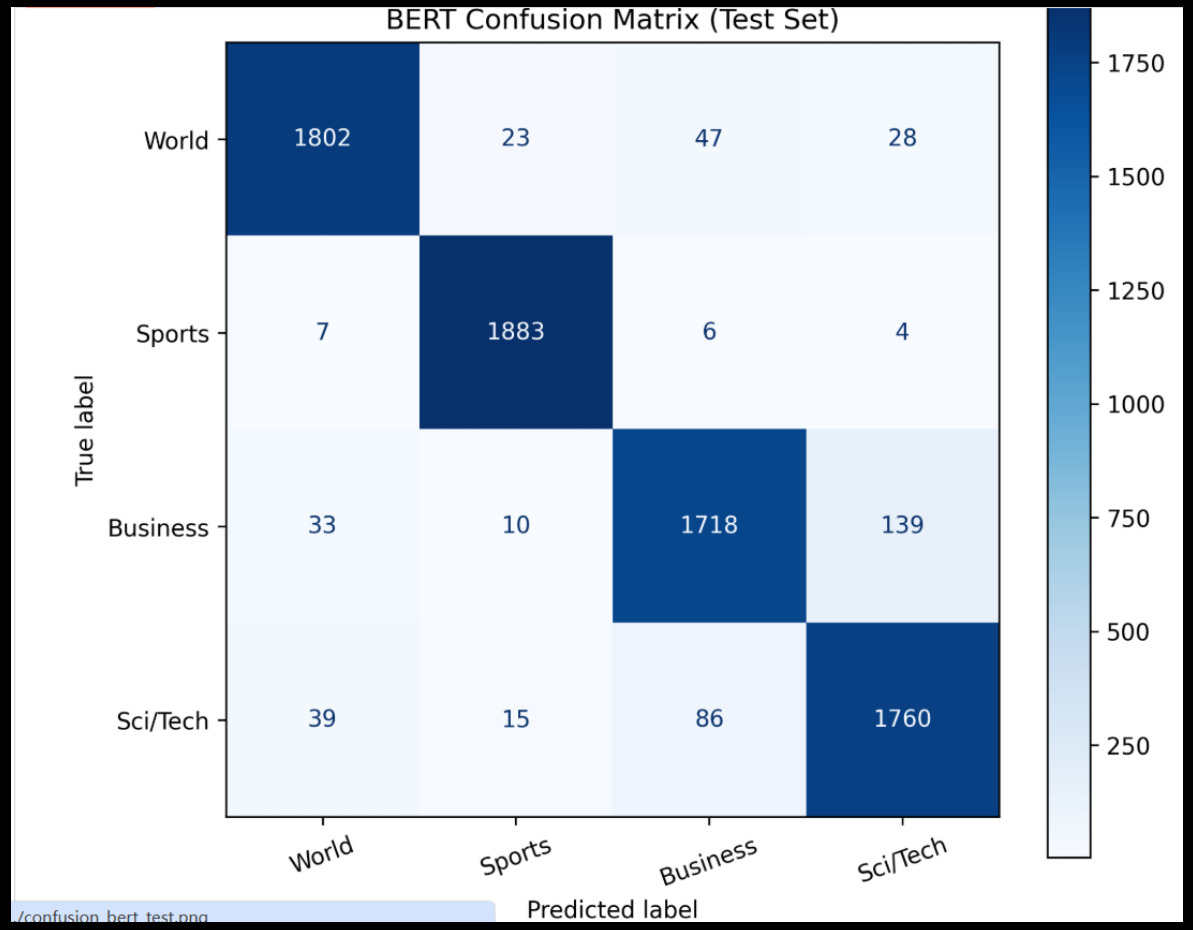
This shows BERT's improvement is real and measurable. It corrected hundreds of mistakes that Logistic Regression made.

Logistic Regression Confusion Matrix



This confusion matrix shows where Logistic Regression makes mistakes. Sports is classified very accurately, but Business and Sci/Tech are often confused with each other. This suggests keyword overlap between these categories creates difficulty for traditional machine learning models.

BERT Confusion Matrix



Compared with Logistic Regression, BERT produces a cleaner confusion matrix with stronger diagonal values, meaning more correct predictions. It especially reduces mistakes in World and Sports categories, and slightly improves Business and Sci/Tech classification. However, Business and Sci/Tech remain the hardest pair even for BERT.

Where BERT Helped

Context Understanding

BERT helped most when:

- categories share vocabulary
- keywords are misleading
- topic meaning depends on surrounding words

Examples:

- Business vs Sci/Tech
- World vs Business

Traditional models often react strongly to keywords. BERT instead understands sentence context, which reduces confusion.

Where BERT Still Struggled

BERT still made errors, especially in categories that naturally overlap.

Both-Wrong Confusion Pair	Count
Business <-> Sci/Tech	164
World <-> Business	62
Sci/Tech <-> World	57

Some AG News categories are genuinely difficult to separate because real news stories often combine technology, business, and world events. For example, a story about a technology company can also be written from a business perspective. Some categories naturally overlap, so even advanced models still make mistakes.

Headline Length Check

- We also checked whether BERT's improvement was mainly related to headline length. The median word count was about 36 for the full test set, BERT-correct examples, BERT-error examples, and Logistic Regression-wrong/BERT-right examples.
- That suggests BERT's improvement was not mainly due to longer or shorter examples. It was more likely related to how the models handled overlapping topic vocabulary.

Main Takeaways

- Logistic Regression was a strong traditional baseline for this task.
- BERT achieved the best overall performance.
- BERT's largest gains came from examples where topic labels had overlapping vocabulary.
- The performance gap was meaningful, but not huge, because TF-IDF-based models already performed well on AG News.
- Compute cost matters: BERT performed better, but it was slower and more expensive to train than the traditional models.

Limitations

- **Single training run:** We report results from one run and did not evaluate variance across multiple random seeds.
- **Limited hyperparameter tuning:** We did not run a large search over BERT settings due to compute and time limits.
- **Clean benchmark dataset:** AG News is balanced and relatively clean, which may not reflect messier real-world news data.
- **Same-distribution evaluation:** The train and test examples come from the same dataset distribution. We did not test on articles from a different source or time period.
- **Short text inputs:** The examples are short headlines or descriptions, so the results may not transfer directly to full-length articles.
- **No calibration analysis:** We evaluated accuracy, precision, recall, and F1, but did not analyze probability calibration.
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Future Work

Possible extensions include:

- Compare BERT with DistilBERT, RoBERTa, or other transformer models
- Run experiments with multiple random seeds
- Evaluate the models on a different news dataset to test domain shift
- Analyze model confidence and calibration
- Build a simple demo for classifying new headlines

Conclusion

- In conclusion, our project showed that while traditional machine learning models like Naive Bayes and Logistic Regression already perform very well on news topic classification, BERT achieved the best overall performance with the highest accuracy and strongest evaluation scores. The main reason for BERT's advantage is its ability to understand words in context rather than relying mainly on keyword matching, which helps reduce mistakes when categories have overlapping vocabulary. However, this improvement also comes with higher computational cost. Overall, our results demonstrate why transformer-based models like BERT have become such an important advancement in Natural Language Processing.